Chp~10
Biomolecules

⇒ Biomolecules may be defined as the complex lifeless organic substances which form the basis of life and are responsible for the growth and maintenance of the living systems.

⇒ They play an active role in the process of metabolism.

⇒ Examples are: Carbohydrates, Proteins, fats and nucleic acids.

CARBOHYDRATES

⇒ Carbohydrates are optically active polyhydroxy aldehydes or polyhydroxy ketones or large polymeric molecules, which on hydrolysis yield polyhydroxy aldehydes or polyhydroxy ketones.

⇒ Examples are: Glucose, fructose, sucrose, starch

⇒ General Formula is $C_x(H_2O)_y$

Classification of Carbohydrates.

1. Monosaccharides: Monosaccharides are those polyhydroxy aldehydes or polyhydroxy ketones which cannot be decomposed further by hydrolysis to give simpler carbohydrates.

=> General Formula is $(CH_2O)_n$.

=> Examples are: Glucose ($C_6H_{12}O_6$)
Fructose ($C_6H_{12}O_6$), galactose ($C_6H_{12}O_6$), ribose ($C_6H_{10}O_5$)

2. Oligosaccharides: Carbohydrates which upon hydrolysis give a definite number (2-10) of molecules of monosaccharides.

=> Types are.
— Disaccharides: Sucrose, Maltose, lactose.
— Trisaccharides: Raffinose

3. Polysaccharides: Carbohydrates which upon hydrolysis give large number of monosaccharide molecules.

=> Examples are: Starch, Cellulose, Glycogen

* Classification on Taste

1. Sugars : Carbohydrates which are sweet in taste are called Sugars.
eg:- Glucose, Fructose, sucrose, lactose.

2. Non- Sugars : Carbohydrates which are not sweet in taste are called non-sugars.
eg:- Starch, cellulose.

* Classification on Reducing Behaviour.

1. Reducing Sugars : Carbohydrates which are capable of reducing Tollens reagent and Fehlings Solution are called reducing sugars.
eg:- All mono and di-saccharides except sucrose.

2. Non- Reducing Sugars : Carbohydrates which are unable to reduce Tollens reagent and Fehlings Solution are non-reducing sugars.
eg:- Sucrose.

GLUCOSE

⇒ Preparation of Glucose

1. From Sucrose

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} \underset{(Glucose)}{C_6H_{12}O_6} + \underset{(Fructose)}{C_6H_{12}O_6}$$

2. From Starch

$$(C_6H_{10}O_5)_n + n\,H_2O \xrightarrow{H^+} n\,C_6H_{12}O_6.$$

⟹ Structure of Glucose

i) Presence of a straight chain of six carbon atoms:

$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow[14H]{HI} \underset{\text{(n-Hexane)}}{CH_3-(CH_2)_4-CH_3} + 6H_2O$$

ii) Presence of a carbonyl group.

— By HCN

$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow{HCN} \begin{array}{c} CH\begin{smallmatrix} OH \\ CN \end{smallmatrix}. \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \\ \text{(A cyanohydrin)} \end{array}$$

— By NH_2OH

$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow[-H_2O]{NH_2OH} \begin{array}{c} CH=NOH \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \\ \text{(An oxime)} \end{array}$$

iii] Presence of an Aldehydic group.

$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow{Br_2\ water} \begin{array}{c} COOH \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array}$$

(Gluconic acid)

iv] Presence of five hydroxyl (-OH) groups

$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow{\begin{array}{c} CH_3CO \\ CH_3CO \end{array} > O} \begin{array}{l} CHO \\ | \\ (CH-O-\overset{O}{\overset{\|}{C}}-CH_3)_4 \\ | \\ CH_2-O-\overset{O}{\overset{\|}{C}}-CH_3 \end{array}$$

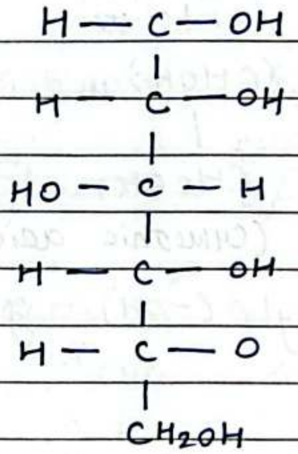
(Pentacetate)

v] Presence of a primary alcohol group.

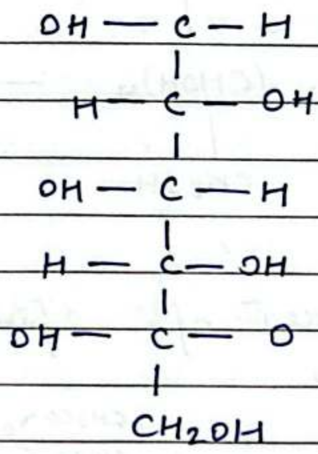
$$\begin{array}{c} CHO \\ | \\ (CHOH)_4 \\ | \\ CH_2OH \end{array} \xrightarrow[HNO_3]{O} \begin{array}{c} COOH \\ | \\ (CHOH)_4 \\ | \\ COOH \end{array}$$

(Saccharic Acid)

=> Structure of Glucose.

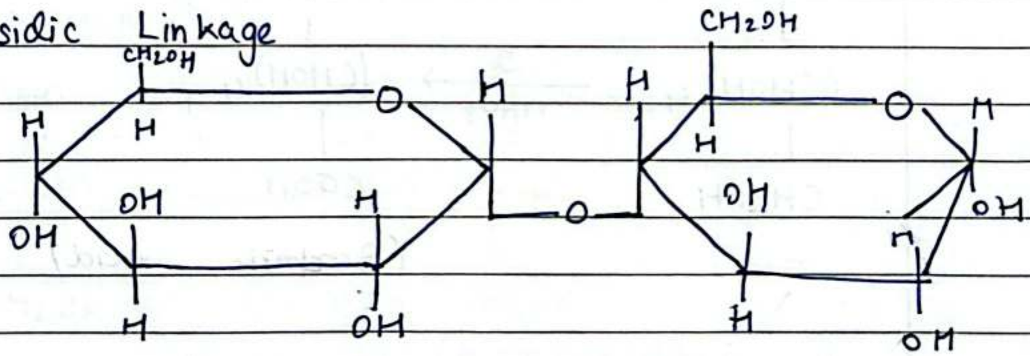


α- Glucose
(when -OH group is towards right)



B- Glucose
(when -OH group is towards left)

=> Glycosidic Linkage



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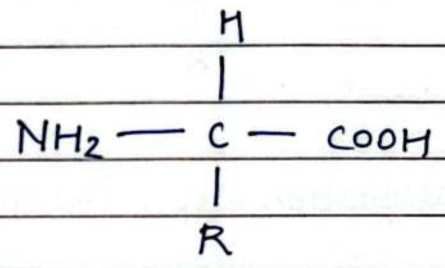
PROTEINS

⇒ Proteins are highly complex nitrogenous organic compound of very high molecular masses All proteins are polymers of α - amino acids.

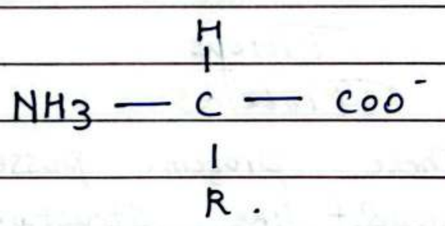
AMINO ACIDS

⇒ Amino acids are the basic building units of Proteins. Their molecules possess both an amino ($-NH_2$) group as well as a carboxylic ($-COOH$) group.

Structure ⇒
of Amino acid



Structure ⇒
of Zwitter
ion



⇒ Classification of Amino Acids.

i) Neutral - when $-NH_2$ group atoms and $-COOH$ group atoms are equal

ii) Acidic - when $-COOH$ group atoms are more than $-NH_2$ groups

iii) Basic - when $-NH_2$ group atoms are more than $-COOH$ group.

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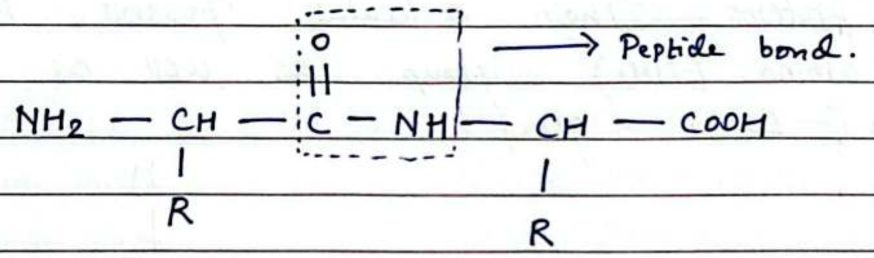
PEPTIDES.

⇒ The compounds obtained by condensation of two or more, α- amino acids are called Peptides.

⇒ Peptide bond is the bond formed by the condensation of two amino acids molecule.

⇒ 

⇒ Classification of Proteins.

Fibrous Proteins	Globular Protins
— These proteins possess thread-like structures.	— These proteins possess folded spheroidal structures
— These proteins are insoluble in water	— These proteins are soluble in water
— These are stable to moderate changes in temperature and pH	— These are very sensitive to changes in temperature and pH.

⇒ Denaturation of Proteins.

The process which leads to a change in physical and biological properties of a Protein without affecting its chemical composition is called denaturation

⇒ Methods.
i] By heat. - Egg.
ii] By Alcohol -
iii] By conc. acids -
iv] By salts of heavy metals.
v] By alkaloidal reagents.
vi] By radiation.

⇒ Examples are : boiling of egg and preparation of cheese from milk.

* NUCLEIC ACIDS

⇒ Nucleic acids constitute an important class of biomolecules which are present in the nuclei of all living cells.

⇒ Nucleic acids are made up of nucleotides, they are actually biopolymers

⇒ Types of Nucleic acids.
i] Deoxyribonucleicacid (DNA)
ii] Ribonucleic acid (RNA)

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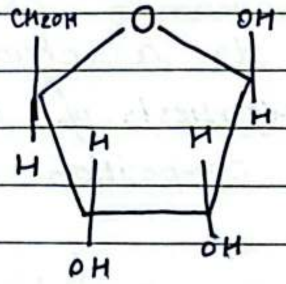
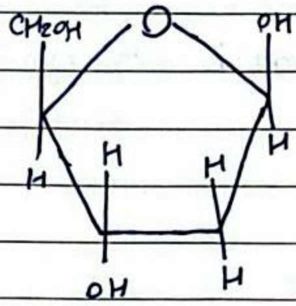
⇒ Structure of Nucleic acids.



⇒ Ribonucleic acid. (RNA)



⇒ Deoxyribonucleic acid (DNA).

⇒ Purines :
Adenine (A) & Guanine (G)

⇒ Pyrimidines :
Thymine (T), Cytosine (C) & Uracil (U)

RNA :	Uracil Adenine Guanine Cytosine
DNA :	Cytosine Adenine Guanine Thymine.

=> NUCLEOSIDES :

BASE + SUGAR = Nucleoside.

=> NUCLEOTIDE :

BASE + SUGAR + PHOSPHATE = Nucleotide.

★ MUTATIONS

=> A chemical change in a DNA molecule which could lead to synthesis of proteins with a different amino acid sequence is termed as mutation.

★ VITAMINS

=> Vitamins may be defined as a group of organic compounds which are required in very small amounts for the healthy growth and normal functioning of animal organisms.

=> i] Fat - Soluble Vitamins : Vitamins A, D, E and K.

ii] Water - Soluble Vitamins : Vitamin B and C.

Vitamin H → exception

Vitamins, Sources and Deficiency Table

	Vitamins	Sources	Deficiency Diseases
1.	Vitamin A	Milk, eggs, fish tomatoes	Night blindness, Xerosis, Xerophthalmia.
2.	Vitamin B_1	milk, eggs, green vegetables	Beri Beri and loss of appetite
3.	Vitamin B_2	Milk, cheese, tomatoes	Cheilosis, Retardation of growth and inflammation of tongue.
4.	Vitamin B_3	Milk, eggs, ~~tomatoes~~	Dermatitis, greying of hair, retardation of body and mental health
5.	Vitamin B_6	Milk, eggs, yeast fish etc.	Anaemia, Insomnia Pellagra, general weakness.
6.	Vitamin B_{12}	Milks, eggs, meat liver etc	Pernicious anaemia Inflammation of tongue and mouth.
7.	Vitamin C	Citrous fruits tomatoes chillies	Scurvy, skin diseases.
8	Vitamin D	milk, eggs, liver	Rickets, deformation of bones and teeth
9.	Vitamin E	Milk, oils, wheat fish.	Sterility, muscular atrophy
10.	Vitamin K	Cabbage, eggs, liver colon	Haemorrhage, lengthening of the time of blood clotting